# Terrain-Aware Uncertainty Quantification and Cross-Sensor Consistency Analysis of Hyperspectral Surface Reflectance

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***Abstract*—Topographic variation introduces substantial uncertainty into hyperspectral surface reflectance (SR) retrieval. This study proposes a combined correction and uncertainty quantification framework that integrates ISOFIT-based atmospheric correction (ATCOR) with a semi-empirical modified Minnaert (MM) topographic correction, while further introducing a novel uncertainty evaluation approach. The framework combines optimal estimation (OE) with a guide to the expression of uncertainty in measurement (GUM)-based uncertainty propagation to explicitly account for terrain parameter uncertainties, enabling efficient and physically consistent characterization of SR uncertainty. Application to the environmental mapping and analysis program (EnMAP) hyperspectral data demonstrates that the method achieves comparable correction accuracy to established algorithms, while providing per-pixel, per-band uncertainty estimates. Validation is performed through cross-sensor comparison with SI-traceable Landsat 8 SR products across diverse land cover types and terrain conditions. The results confirm the reliability of the proposed uncertainty estimates and highlight their value for assessing cross-sensor reflectance consistency. Analysis further reveals that reflectance uncertainty generally follows a Gaussian distribution and that slope uncertainty is the dominant driver of reflectance uncertainty. Overall, this work delivers a practical framework for uncertainty-aware SR retrieval in mountainous regions and establishes a pathway toward uncertainty-based interoperability of hyperspectral products across multiple sensors.**

***Index Terms*—Consistency analysis, hyperspectral remote sensing, topographic correction, uncertainty quantification.**

## I. Introduction

With the increasing number of spaceborne hyperspectral missions and onboard instruments now in orbit, the accuracy, consistency, and interoperability of hyperspectral surface reflectance (SR) products have become frontier issues of significant concern. Because of their high spectral resolution and often finer spatial resolution, hyperspectral sensors typically capture weak radiometric signals per spectral band, which are highly susceptible to complex interactions among atmospheric conditions, terrain variations, and sensor characteristics. These interactions introduce systematic and random errors that contribute to uncertainty in the reflectance products. Such uncertainties may propagate through subsequent processing steps, affecting the reliability of downstream applications. Consequently, the hyperspectral remote sensing community has recognized the urgent need for rigorous and traceable quantification of reflectance uncertainty to support product quality assessment and robust application performance.

Initiatives such as the quality assurance framework for Earth observation (QA4EO) and fidelity and uncertainty in climate data records from earth observations (FIDUCEOs) have laid out high-level frameworks to support rigorous uncertainty analysis in satellite-derived products [1], [2]. These frameworks emphasize the provision of per-pixel uncertainty estimates alongside data products, which not only facilitate the optimization of retrieval algorithms but also enable uncertainty propagation through downstream processing chains and support informed end-user decision-making. Building on this, the European Space Agency (ESA) has launched a series of fiducial reference measurement (FRM) projects [3], which shift the validation paradigm away from relying solely on simultaneous in situ measurements. Instead, these efforts emphasize metrological traceability and independent quality evaluation of ground-based observations. Based on the FRM framework, the National Physical Laboratory (NPL), U.K. developed a Monte Carlo–based methodology to estimate SR uncertainty from satellite data processing chains, incorporating both spaceborne and ground-based error sources [4]. Furthermore, they produce spatially explicit, high-resolution reference maps with per-pixel uncertainty estimates applicable to a variety of biogeophysical variables, including leaf area index (LAI), fractional vegetation cover (FCOVER), and canopy water content (CWC) [5].

At the mission level, the Sentinel-2 program has served as a milestone for operational uncertainty quantification. Extensive studies have investigated uncertainty in Level-1C radiance products [6], [7], Level-2A SR products [8], and derived land surface phenology indicators [9], all assessed within a Monte Carlo framework. More recently, uncertainty-aware land cover classification approaches have been developed, explicitly incorporating uncertainty into classification schemes [10], [11].

Beyond Monte Carlo methods, alternative approaches based on optimal estimation (OE) have been advanced. Researchers

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at NASA Jet Propulsion Laboratory (JPL) have demonstrated the effectiveness of OE for atmospheric correction (ATCOR), using the diagonal elements of the posterior covariance matrix to quantify reflectance uncertainty [12], [13]. This approach has further been extended to hyperspectral data from the Earth surface mineral dust source investigation (EMIT) mission, with retrieved reflectance and its associated uncertainties validated against homogeneous standard field networks [14], [15]. In parallel, uncertainty estimation has also been increasingly incorporated into downstream biophysical variable retrieval [16].

In rugged terrain, however, topography exerts a strong influence on illumination geometry. Variations in slope and aspect create brightness differences for identical land cover types, thereby introducing additional sources of uncertainty in SR products. Considerable progress has been made in recent years to mitigate these effects. Experimental evaluations of topographic correction approaches have shown that combining a physical model (Lambertian, LA) with an empirical method (Statistical-Empirical, SE)—the so-called LA + SE approach—as well as the modified Minnaert (MM) method, provides the most reliable performance under satellite observation conditions; for airborne hyperspectral data, the MM method has proven effective both statistically and visually [17]. In the United States, an integrated retrieval framework that couples atmospheric, terrain, and SR corrections within an OE framework has been proposed. Tests with airborne visible/infrared imaging spectrometer (AVIRIS) airborne data confirmed its ability to reduce intrascene spectral inconsistencies and enhance retrieval robustness [18]. In addition, comprehensive reviews and recent advances in topographic correction methods have been reported, providing systematic summaries of existing approaches for optical satellite data [19], [20].

With respect to uncertainty analysis that explicitly accounts for terrain effects, Ma et al. [21] examined the uncertainty propagation chains of two semi-empirical topographic correction models and proposed strategies to minimize uncertainty introduced during topographic adjustments. Other researchers have pointed out that global digital elevation model (DEM) products exhibit significant errors compared with airborne LiDAR, especially in slope and aspect estimation, which in turn affect the computation of the cosine of the solar incidence angle ($\beta$) [22]. Simulated analyses have demonstrated how DEM data uncertainty propagates into slope, aspect, and cosine of solar incidence angle [23].

Nevertheless, existing research has not adequately addressed how terrain variability contributes to reflectance uncertainty quantification. Previous studies have primarily focused on ATCOR–related uncertainty propagation. For instance, Thompson et al. [13] and Origo et al. [4] analyzed uncertainty estimation within the frameworks of OE and GUM, respectively, but without explicitly accounting for the role of terrain variability. Similarly, Carmon et al. [23] investigated the influence of topographic illumination geometry yet did not propagate these effects through to SR uncertainty. Consequently, the impacts of terrain-related parameters—such as slope, aspect, and solar incidence angle—on SR uncertainty remain insufficiently characterized. This gap gives rise to two major challenges. First, neglecting the role of terrain variability in reflectance retrieval introduces additional uncertainty, which not only compromises the reliability of uncertainty assessments but also undermines sensor-to-sensor interoperability and constrains the broader applicability of uncertainty information. Second, current reflectance uncertainty estimates have not yet undergone systematic validation, which further limits their operational implementation and hinders the transition of uncertainty information into standardized data products.

To overcome these limitations, a method is proposed that combines OE with the MM method, thereby explicitly propagating terrain parameter uncertainties into SR uncertainties. Rather than treating slope, aspect, and solar incidence angle as fixed inputs, the framework quantifies their variability and evaluates how these uncertainties affect retrieved reflectance. By embedding this treatment within a GUM-based uncertainty propagation scheme, the method delivers per-pixel, per-band uncertainty estimates that are physically consistent and directly linked to terrain variability. In addition, a quantitative cross-sensor validation strategy is introduced to verify the reliability of the derived uncertainty estimates using independently calibrated reference data. This dual-level innovation—linking terrain-aware uncertainty propagation with cross-sensor uncertainty validation—establishes a rigorous and traceable foundation for reliable hyperspectral SR retrieval in rugged terrain.

This article is organized as follows. Section II presents the atmospheric and topographic correction method, the uncertainty quantification framework, and the validation of the methodology. Section III describes the satellite hyperspectral datasets and the associated digital surface model (DSM) data used in the analysis, and reports the results of reflectance retrieval and corresponding uncertainty for selected two spaceborne scenes. Section IV provides a discussion of the findings, and conclusions are drawn in Section V.

## II. Methods

The methodology of this study is designed to quantitatively evaluate SR uncertainty in rugged terrain and to assess its implications for multisensor consistency analysis. To ensure accurate reflectance retrievals, ISOFIT-based ATCOR is first applied (Section II-A). Section II-B establishes the theoretical compatibility between the OE and GUM-based uncertainty quantification frameworks, providing the metrological foundation for the subsequent uncertainty propagation. Section II-C introduces MM topographic correction model, which serves as the basis for terrain effect compensation. Building on these corrections, an uncertainty quantification methodology is developed by combining OE with the GUM, explicitly incorporating terrain parameter uncertainties into the reflectance uncertainty budget (Section II-D). Finally, the reliability and applicability of the uncertainty estimates are evaluated through intersensor comparisons with Landsat data, providing a practical demonstration of uncertainty-aware multimission reflectance consistency analysis (Section II-E). The methodological framework of this study is summarized in Fig. 1.

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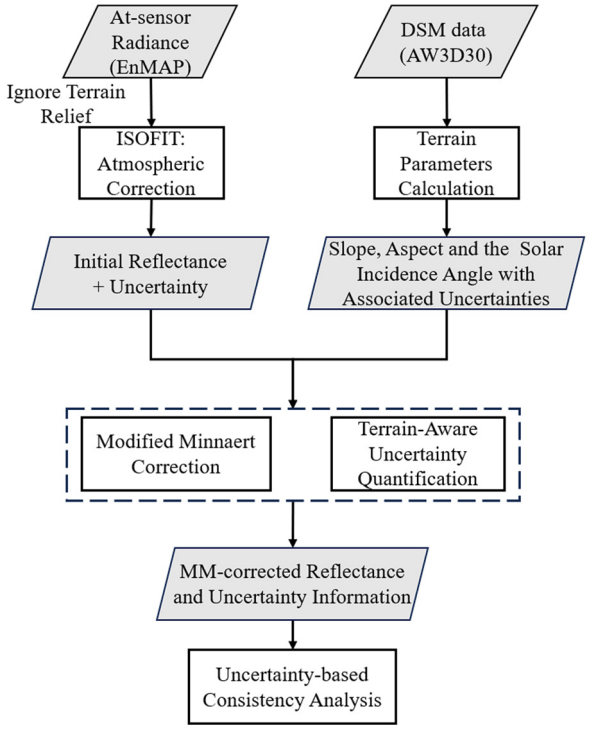



Fig. 1. Schematic workflow of the method.

### A. ATCOR and Its Uncertainties

SR is retrieved through ATCOR using imaging spectrometer optimal fitting (ISOFIT) version 3 [24]. ISOFIT estimates SR across $n$ spectral channels along with atmospheric state variables such as water vapor and aerosol optical depth from $n$ observed radiance channels. Since the number of unknowns in hyperspectral retrieval typically exceeds the number of observations, this ill-posed inverse problem requires additional constraints for a physically meaningful solution.

To address this, the retrieval is performed using the OE method [25], a widely adopted technique in atmospheric remote sensing that incorporates a statistically defined prior to regularize the inversion. In this study, the prior imposes weak constraints on the solution space, effectively describing plausible spectral shapes within relevant wavelength ranges while avoiding excessive bias.

During the OE-based ATCOR process, the radiative transfer forward model is simulated using a hybrid surrogate radiative transfer model/neural network (sRTMnet) emulator [26], a machine-learning emulator trained on moderate resolution atmospheric transmission (MODTRAN) simulations [27].

The forward model is expressed as follows:

$$y = f(x) + \varepsilon = Kx + \varepsilon \tag{1}$$

where $y$ denotes the observed at-sensor radiance, $\varepsilon$ represents random errors, including instrument noise and forward model inaccuracies. The Jacobian matrix $K$ quantifies the sensitivity of the forward model $f(x)$ to the state $x$.

Within the OE framework, the retrieval seeks the state vector that minimizes a cost function $\chi^2$ in the following equation, which balances deviations from the prior distribution and mismatches between modeled and observed radiance

$$\chi^2 = \frac{1}{2}(x - x_a)^T S_a^{-1}(x - x_a) + \frac{1}{2}(y - f(x))^T S_\varepsilon^{-1}(y - f(x)) \tag{2}$$

where $x_a$ is the prior mean of the state vector, $S_a$ is the prior covariance matrix, and $S_\varepsilon$ is the total observation error covariance matrix. The latter includes the sensor noise covariance matrix $S_y$ and unmodeled parameters $S_b$. Sensor noise is estimated from the instrument signal-to-noise ratio (SNR). The first term in (2) penalizes the state $x$ deviation from the prior $x_a$, while the second term accounts for the mismatch between the forward model prediction and the measured at-sensor radiance.

By rearranging (1), (3) is obtained, and the gradient of the cost function can be expressed as shown in (4)

$$K = \partial y/\partial x = \partial f(x)/\partial x \tag{3}$$

$$\nabla\chi^2(x) = -K^T S_\varepsilon^{-1}(y - f(x)) + S_a^{-1}(x - x_a). \tag{4}$$

An iterative optimization approach is employed to minimize the cost function. At each iteration, the Jacobian matrix $K$ is recomputed, and the state vector is updated using (5), thereby avoiding explicit gradient computation. The process continues until a predefined convergence threshold—typically a 1% change in the state vector—or a maximum number of iterations is reached, at which point the optimal state estimate is obtained

$$\hat{x}_{i+1} = \hat{x}_i + \left(S_a^{-1} + K_i^T S_\varepsilon^{-1} K_i\right)^{-1} \left[K_i^T S_\varepsilon^{-1}\left[y - f(\hat{x}_i)\right] - S_a^{-1}(\hat{x}_i - x_a)\right]. \tag{5}$$

The posterior predicted distribution has covariance $\hat{S}$ given by the following equation:

$$\hat{S} = \left(K^T S_\varepsilon^{-1} K + S_a^{-1}\right)^{-1}. \tag{6}$$

This matrix quantifies the uncertainty introduced by random noise and forward model imperfections during the retrieval process. The diagonal elements represent the variances of the state vector elements, from which standard deviations (i.e., uncertainties) can be derived. The off-diagonal terms describe the spectral correlation between reflectance estimates at different wavelengths. Consequently, this iterative optimization framework enables pixel-wise and band-wise quantification of SR uncertainty [28]. Importantly, obtaining accurate reflectance uncertainty requires not only a realistic characterization of sensor-specific noise, but also careful specification of atmospheric parameters and the prior covariance matrix of SR.

Together, the OE method searches for the state vector that maximizes the posterior probability given the observed radiance, the forward model, and the prior constraints. This probabilistic formulation not only stabilizes the inversion but also provides a natural quantification of retrieval uncertainties through the posterior covariance matrix.

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### B. Compatibility Between the OE and GUM-Based Uncertainty Quantification

The compatibility between OE-based and GUM-based uncertainty evaluation has been extensively discussed in previous metrological studies, which provide both theoretical and numerical analyses of their equivalence and applicability in measurement science [29], [30]. Building upon these findings, the present study further derives the theoretical relationship between the two frameworks in the context of ATCOR.

The ATCOR process can be expressed as a forward radiative transfer model linking the measured at-sensor radiance $y$ to the SR $\rho_t$, path reflectance $\rho_a$, spherical albedo $s$, and atmospheric transmission $t$. Following [12], the relationship between the apparent top-of-atmosphere (TOA) reflectance $\rho_{\text{obs}}$ and the SR $\rho_s$ can be written as follows:

$$\rho_{\text{obs}} = \rho_a + \frac{t \cdot \rho_t}{1 - s \cdot \rho_t}. \tag{7}$$

For inversion, this model can be reformulated as follows:

$$\rho_t = \frac{\rho_{\text{obs}} - \rho_a}{t + s\,(\rho_{\text{obs}} - \rho_a)}. \tag{8}$$

This analytical expression provides an explicit link between the variables ($\rho_{\text{obs}}$, $\rho_a$, $t$, $s$) and the desired measurand $\rho_t$, enabling uncertainty propagation analysis under both GUM and OE frameworks.

According to the GUM, the variance–covariance matrix of the output measurand $S_{\rho_t}^{\text{GUM}}$ can be obtained through the law of propagation of uncertainty (LPU) [1]

$$S_{\rho_t}^{\text{GUM}} = J S_{\varepsilon} J^T \tag{9}$$

where $S_{\varepsilon}$ denotes the total observation error covariance matrix, consistent with the definition in (2). And $J$ is the Jacobian matrix composed of partial derivatives

$$J = \left[\frac{\partial \rho_t}{\partial \rho_{\text{obs}}}, \frac{\partial \rho_t}{\partial \rho_a}, \frac{\partial \rho_t}{\partial t}, \frac{\partial \rho_t}{\partial s}\right]. \tag{10}$$

Under the assumption of independent variables, $S_{\varepsilon}$ becomes a diagonal matrix containing the variances of $\rho_{\text{obs}}$, $\rho_a$, $t$ and $s$. Equation (9) ensures that the propagated reflectance uncertainty is traceable to the defined input uncertainties of the radiative transfer model, following metrological principles.

Comparing (6)–(9) demonstrates their conceptual equivalence: both represent the dispersion of possible values that can reasonably be attributed to the measurand $\rho_t$. The OE posterior covariance incorporates the same measurement model as GUM but adds prior uncertainty $S_a$, thereby generalizing the GUM propagation into a Bayesian context. Under locally linear conditions, the forward relation can be expressed as $y = f(x) \approx Kx$ and its inverse as $x \approx Jy$. In this linearized setting, $J$ and $K$ are matrix inverses ($K \approx J^{-1}$), so the relationship between the two formulations is shown in the following equation:

$$\hat{S} = S_{\rho_t}^{\text{OE}} = \left(\left(S_{\rho_t}^{\text{GUM}}\right)^{-1} + S_a^{-1}\right)^{-1}. \tag{11}$$

Equation (11) ensures that the OE-derived uncertainty remains GUM-compatible while integrating prior information.

This theoretical equivalence establishes that the OE posterior covariance can be interpreted as a GUM-consistent standard uncertainty of the retrieved SR. Consequently, the framework integrates OE-based uncertainty from ATCOR with GUM-based uncertainty propagation from terrain correction within a unified metrological foundation. This compatibility provides a solid theoretical basis for subsequently applying the GUM approach to quantify the uncertainty of topographic correction.

### C. Topographic Correction Method

Topographic correction commonly relies on a DEM or DSM to account for local variations in illumination geometry. Given the geographic coordinates and acquisition time of the scene, the solar zenith angle $\theta_s$ and solar azimuth angle $\phi_s$ can be derived. Using slope $S$ and aspect $A$ information from the terrain, the local solar illumination angle $\beta$—i.e., the angle between the surface normal and the direction of solar illumination—can then be computed for each pixel [31]

$$\cos\beta = \cos S \cos\theta_s + \sin S \sin\theta_s \cos(A - \phi_s). \tag{12}$$

Among various approaches, the MM method is a widely used semi-empirical topographic correction approach designed to reduce the effects of topographic variations on reflectance by adjusting for changes in illumination geometry [32], [33]. Its accuracy and robustness have been demonstrated across diverse terrain and land cover conditions [34]. In rugged terrain, the solar incidence angle $\beta$ is considered the dominant factor, while viewing geometry is often neglected.

As an initial step, the SR $\rho_L$ is modeled under the Lambertian assumption

$$\rho_L = \rho_t \frac{\cos\theta_s}{\cos\beta} \tag{13}$$

where $\rho_t$ is the target SR, which in this study is obtained from the OE retrieval results.

Then, if the solar incidence angle $\beta$ exceeds a threshold incidence angle $\beta_T$ (i.e., in areas of low illumination, where $\rho_L$ might become high), the updated SR is reduced according to the following equation:

$$\rho_{\text{MM}} = \rho_L \left(\frac{\cos\beta}{\cos\beta_T}\right)^b \tag{14}$$

where $\rho_{\text{MM}}$ denotes the MM-corrected SR and $b$ is an empirical exponent dependent on surface type. For clarity, the latter term is defined as the geometric function $G$

$$G = (\cos\beta_T / \cos\beta)^b. \tag{15}$$

A coarse classification distinguishes vegetated from nonvegetated pixels. Vegetation is identified using a reflectance index threshold: $\rho_{850\,\text{nm}}/\rho_{660\,\text{nm}} > 3$. For nonvegetated pixels, a wavelength-independent exponent $b = 1/2$ is used. For vegetated pixels, based on extensive testing across mountainous regions with varying vegetation cover, the exponent $b$ is defined with spectral dependence as follows.

1) $b = 3/4$ for bands with $\lambda < 720$ nm.
2) $b = 1/3$ for bands with $\lambda \geq 720$ nm.

The threshold incidence angle $\beta_T$ for applying MM correction can be user-defined. Recommended empirical rules for assigning this threshold are:

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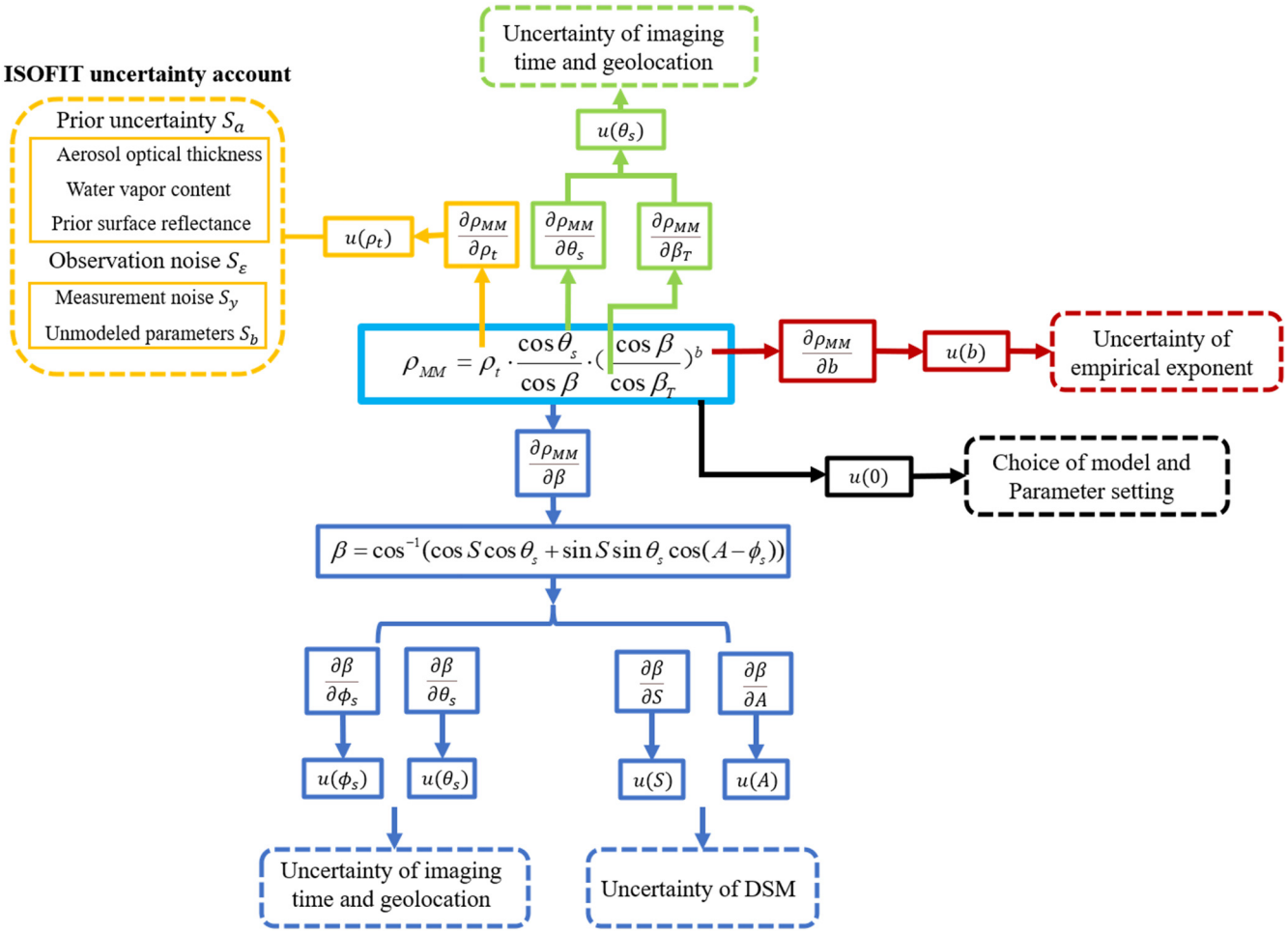



Fig. 2. Uncertainty tree diagram for the MM-corrected SR.

1) If $\theta_s < 35°$ then $\beta_T = \theta_s + 25°$.
2) If $35° \le \theta_s < 45°$ then $\beta_T = \theta_s + 20°$.
3) If $45 \le \theta_s < 50°$ then $\beta_T = \theta_s + 15°$.
4) If $50 \le \theta_s \le 60°$ then $\beta_T = \theta_s + 10°$.
5) If $\theta_s > 60°$ then $\beta_T = \max(70°, \theta_s + 5°)$.

To avoid under- or over-correction of SR, the geometric function $G$ is constrained within predefined limits. Specifically, values of $G$ greater than 1 are set to 1, while values below a user-defined lower threshold (defaulting to $G = 0.25$) are reset to that threshold.

Accordingly, the MM method is applied as follows:

$$\rho_{\mathrm{MM}} = \rho_t \cdot \frac{\cos\theta_s}{\cos\beta} \cdot \left(\frac{\cos\beta}{\cos\beta_T}\right)^b. \tag{16}$$

Nevertheless, residual uncertainties persist, primarily arising from noise in the DEM/DSM used to compute illumination geometry and from sensor radiometric noise. A quantitative assessment of these uncertainties is therefore required, motivating the uncertainty quantification framework introduced in Section II-D.

### *D. Terrain-Aware Uncertainty Quantification*

Uncertainties from DEM/DSM are particularly critical in rugged terrain, where in situ reference measurements are often inaccessible. This makes it essential to perform rigorous uncertainty quantification of SR products, ensuring their reliability in regions where direct validation is not feasible. To address this, the framework combines OE with the GUM-based uncertainty evaluation approach, enabling traceable and physically consistent quantification of terrain-induced reflectance uncertainties.

The GUM is the fundamental metrological standard for uncertainty evaluation and has been extended to a broad range of scientific measurement applications [35], [36]. Building on these principles, the Committee on Earth Observation Satellites (CEOSs) introduced the QA4EO, which emphasizes traceability, quantitative uncertainty reporting, and the use of the LPU for Earth observation products [1]. This ensures that data values and their uncertainties are linked through an unbroken chain of comparisons to recognized standards.

OE, in contrast, provides a probabilistic framework for ATCOR by combining prior information with satellite observations. The posterior probability distribution represents the updated state estimate, and its standard deviation directly quantifies the retrieval uncertainty [37]. Importantly, both GUM and OE share a consistent definition of standard uncertainty—expressed as a parameter that characterizes the dispersion of the values that could reasonably be attributed

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to the measurand. From both a conceptual standpoint and the theoretical derivations presented in Section II-B, this alignment underpins their integration within the method, enabling GUM-compatible uncertainty estimates that incorporate OE-derived posterior covariance information.

For the method that combines GUM and OE, the tree diagram and uncertainty sources are as shown in Fig. 2. The sources of uncertainty include

1) *Initial SR $\rho_t$:* This component of uncertainty is derived from the ISOFIT retrieval and consists of the following elements.
   a) *Prior Uncertainties*, including uncertainty in prior SR, aerosol optical thickness, and water vapor content.
   b) *Observation Noise $S_\varepsilon$*, including instrument measurement noise and model residuals due to unmodeled physical processes. Detailed information can be found in [12]. Based on these inputs, ISOFIT yields the posterior covariance matrix $\hat{S}$, the square root of its diagonal elements can be interpreted as the uncertainty $u(\rho_t)$.
2) *Imaging Geometry:* This includes the uncertainties in solar zenith and azimuth angles, which originate from uncertainties in image acquisition time and geolocation. However, compared to other uncertainty sources, their impact is negligible and can be disregarded.
3) *Empirical Coefficient b:* The coefficient $b$ in the MM correction is successfully tested on many rugged terrain scenes and has been widely adopted in previous studies. Given its stability and the strong empirical basis, the uncertainty associated with $b$ is considered negligible compared with other uncertainty sources and is, therefore, not propagated in the uncertainty analysis.
4) *Choice of Model and Parameter Setting [$u(0)$]:* The MM method is a semi-empirical topographic correction model and, as such, inherently carries model-related uncertainties. In particular, the empirical determination of the $\beta_T$ parameter introduces additional uncertainty, as it is often scene-dependent and sensitive to local illumination and surface conditions. Although the MM method partially accounts for the bidirectional reflectance distribution function (BRDF) behavior through the incorporation of the incidence-angle term, it does not fully resolve the anisotropic reflectance characteristics of complex surfaces. Consequently, both the empirical formulation of the MM method and the approximation of BRDF effects contribute to an overall model-based uncertainty component, denoted as $u(0)$.
   A rigorous investigation of these model-inherent uncertainties lies beyond the scope of the present study and will be systematically examined in future work.
5) *Terrain Parameters:* These uncertainties originate from the DSM data, which include both vertical and horizontal uncertainty components. The uncertainties in slope $S$, aspect $A$, and the cosine of the solar incidence angle $\cos\beta$ are calculated using the LPU. Specifically, $f_x$ and $f_y$ represent the elevation gradients in the east-west and north-south directions. $Z_i$ $(i = 1, 2, \ldots, 9)$ denotes the elevation values of the $3 \times 3$ neighborhood grid centered on the target pixel. The grid resolution is denoted by $g$. The specific equations for calculation and uncertainty propagation are as follows.

$f_x$, $f_y$ and their uncertainties are calculated as follows (17) and (18), shown at the bottom of the next page, where $u(Z_i)$ and $u(Z_j)$ represent the standard uncertainties of the $Z_i$ and $Z_j$, respectively. $r_{ij}$ is the correlation coefficient between the elevation $Z_i$ and $Z_j$.

Furthermore, the calculations of $S$ and $A$, as well as their associated uncertainties, are shown in the following equations:

$$\begin{cases} S = \arctan\left(\sqrt{f_x^2 + f_y^2}\right) \\ A = 270^\circ + \arctan\left(f_y / f_x\right) - 90^\circ f_x / |f_x| \end{cases} \tag{19}$$

$$\begin{cases} u(S) = \sqrt{\left(\dfrac{\partial S}{\partial f_x} u(f_x)\right)^2 + \left(\dfrac{\partial S}{\partial f_x} u(f_y)\right)^2} \\ u(A) = \sqrt{\left(\dfrac{\partial A}{\partial f_x} u(f_x)\right)^2 + \left(\dfrac{\partial A}{\partial f_x} u(f_y)\right)^2}. \end{cases} \tag{20}$$

The uncertainty associated with the cosine of the solar incidence angle can be obtained by jointly applying (12), (19), and (20)

$$u(\cos\beta) = \sqrt{\begin{aligned} &\left(\frac{\partial \cos\beta}{\partial S} u(S)\right)^2 + \left(\frac{\partial \cos\beta}{\partial A} u(A)\right)^2 \\ &+ 2 \frac{\partial \cos\beta}{\partial S} \frac{\partial \cos\beta}{\partial A} r_{S,A} u(S) u(A) \end{aligned}} \tag{21}$$

where $r_{S,A}$ is the correlation coefficient between the $S$ and $A$.

At this point, all sources of uncertainty can be traced and quantified. Based on the core MM correction formula [(16)] and applying the LPU from the GUM framework, while neglecting the uncertainties associated with solar zenith angle and the empirical coefficient $b$, the uncertainty of the terrain-corrected SR can be derived as follows:

$$\begin{aligned} u^2(\rho_{\mathrm{MM}}) = &\left(\frac{\partial \rho_{\mathrm{MM}}}{\partial \rho_t} u(\rho_t)\right)^2 + \left(\frac{\partial \rho_{\mathrm{MM}}}{\partial \beta} u(\beta)\right)^2 \\ &+ 2 \frac{\partial \rho_{\mathrm{MM}}}{\partial \rho_t} \frac{\partial \rho_{\mathrm{MM}}}{\partial \beta} r_{\rho_t,\beta} u(\rho_t) u(\beta) \end{aligned} \tag{22}$$

where $r_{\rho_t,\beta}$ is the correlation coefficient between $\beta$ and $\rho_t$. This correlation is empirically estimated for each spectral band using the Pearson correlation coefficient [38], capturing the wavelength-dependent coupling between illumination geometry and SR variability. The partial derivatives in (22) are analytically derived from the MM correction function to ensure computational stability and physical interpretability. In this study, only per-band variance components are propagated to focus on dominant uncertainty sources, while the potential effects of interband covariance are left for future investigation.

In summary, the framework for evaluating SR uncertainty in rugged terrain consists of three steps, as follows:

1) Quantify uncertainties of traceable input variables and define their probability distributions.
2) Use OE retrieval to obtain the posterior covariance matrix, where diagonal terms provide per-band uncertainties.

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3) Propagate DSM-derived terrain parameter uncertainties through the MM correction using the LPU, rigorously accounting for correlations.

This combined approach enables systematic quantification of reflectance uncertainties in rugged terrain, ensuring traceability to metrological standards while capturing the probabilistic nature of ATCOR.

### *E. Validation of Methodology*

This study uses hyperspectral data acquired by environmental mapping and analysis program (EnMAP) [39], a spaceborne imaging spectrometer developed by the German Aerospace Center (DLR). EnMAP provides high-quality SR data with 30 m spatial resolution and 224 contiguous spectral bands covering the 420~2450 nm range. The sensor is designed for precise environmental monitoring, offering low noise levels and accurate radiometric calibration.

The accompanying terrain data is chosen from ALOS Global DSM "ALOS World 3-D–30 m" (AW3D30), with an elevation accuracy of 5 m and a horizontal accuracy of 5 m, evaluating in terms of root mean square error (RMSE) [40]. This level of accuracy ensures its suitability for supporting topographic correction and subsequent uncertainty propagation in rugged mountainous terrain.

The performance of the ISOFIT + MM correction workflow is first validated. Since obtaining synchronous in situ SR measurements in rugged mountainous terrain is extremely challenging, the correction performance is instead evaluated using indirect methods. These include 1) visual inspection of spatial patterns and 2) correlation analysis between SR and illumination geometry [41].

Beyond conventional correction validation, the primary innovation of this work is the quantitative validation of uncertainty estimates through cross-sensor comparison. For this purpose, hyperspectral EnMAP-derived reflectance is rigorously compared against Landsat 8 SR products, which are internationally recognized as traceable to SI standards and are characterized by an established uncertainty budget of $0.05\rho + 0.005$ [42], [43]. To enable spectral consistency between the two sensors, the EnMAP hyperspectral reflectance is spectrally resampled using the published spectral response functions (SRFs) of Landsat 8 [44], following established cross-sensor comparison practices [45].

For a complementary quantitative assessment, the Bias, RMSE [42] and Pearson correlation coefficient ($R^2$) [38] are computed between EnMAP and Landsat SR for both the NC case and the AC + TC case over the selected typical validation sites

$$\text{Bias} = \frac{1}{N}\sum_{i=1}^{N}\left|\rho_{1,i} - \rho_{2,i}\right| \tag{23}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\rho_{1,i} - \rho_{2,i})^2} \tag{24}$$

$$R^2 = \left(\frac{\text{Cov}(\rho_1, \rho_2)}{\sigma_{\rho_1} \times \sigma_{\rho_2}}\right)^2 \tag{25}$$

where $N$ is the number of compared bands. $\rho_{1,i}$ and $\rho_{2,i}$ are the SR values from the two sensors at the $i$th spectral band, respectively. $\text{Cov}(\rho_1, \rho_2)$ represents the covariance between the two reflectance datasets. $\sigma_{\rho_1}$ and $\sigma_{\rho_2}$ are their respective standard deviations.

Typical validation sites are then analyzed by assessing the consistency of uncertainty estimates across multiple sensor platforms [46], [47]. This is done using the $E_n$-ratio, defined as follows:

$$E_N = \frac{|\rho_1 - \rho_2|}{k \cdot \sqrt{u_1^2 + u_2^2}} \tag{26}$$

where $\rho_1$ and $\rho_2$ are the SR values from the two sensors, and $u_1$ and $u_2$ are their respective uncertainties. The factor $k$ represents the coverage factor. Here, the uncertainties are assumed to follow a Gaussian distribution with a coverage factor of $k = 1.96$ (95% confidence level), while further discussion on the assumption of uncertainty distributions is provided in Section IV-A. Conceptually, the $E_n$-ratio measures whether differences between two independent reflectance estimates remain statistically consistent within their combined uncertainty bounds. Under the assumption of Gaussian distributed uncertainties, the $E_n$-ratio is essentially equivalent to a statistical test: an $E_n$-ratio $\leq 1$ indicates that the discrepancy between two datasets lies within the uncertainty envelope and thus they can be regarded as statistically consistent, whereas

$$\begin{cases} f_x = [Z_7 - Z_1 + 2(Z_8 - Z_2) + Z_9 - Z_3]/8g \\ f_y = [Z_3 - Z_1 + 2(Z_6 - Z_4) + Z_9 - Z_7]/8g \end{cases} \tag{17}$$

$$\begin{cases} u(f_x) = \sqrt{\sum_{i=1,2,3,7,8,9}\left(\frac{\partial f_x}{\partial Z_i}\right)^2 u^2(Z_i) + \left(\frac{\partial f_x}{\partial g}\right)^2 u^2(g) + 2\sum_{i=1,2,3,7,8,9}\sum_{j=1,2,3,7,8,9,j>i}\frac{\partial f_x}{\partial Z_i}\frac{\partial f_x}{\partial Z_j} r_{ij} u(Z_i)\, u(Z_j)} \\ u(f_y) = \sqrt{\sum_{i=1,3,4,6,7,9}\left(\frac{\partial f_y}{\partial Z_i}\right)^2 u^2(Z_i) + \left(\frac{\partial f_y}{\partial g}\right)^2 u^2(g) + 2\sum_{i=1,3,4,6,7,9}\sum_{j=1,3,4,6,7,9,j>i}\frac{\partial f_y}{\partial Z_i}\frac{\partial f_y}{\partial Z_j} r_{ij} u(Z_i)\, u(Z_j)} \end{cases} \tag{18}$$

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an $E_n$-ratio > 1 indicates that the discrepancy exceeds the combined uncertainty bounds, signaling inconsistency. If the uncertainty budgets are incomplete or unrealistic (e.g., neglecting terrain-related contributions), the combined uncertainty will be underestimated, the denominator reduced, and the $E_n$-ratio artificially inflated, which may lead to misleading conclusions on interoperability. In contrast, when the uncertainty evaluation is comprehensive and accurate, the $E_n$-ratio provides a properly normalized measure that faithfully reflects the level of consistency between the two sensor datasets. Therefore, only when supported by robust uncertainty characterization does an $E_n$-ratio below unity provide credible evidence for interoperability between sensors [46].

By directly embedding uncertainty into the cross-sensor evaluation, this framework goes beyond traditional reflectance intercomparisons. It provides a quantitative, uncertainty-aware basis for assessing multimission radiometric consistency, thereby addressing a critical need for upcoming hyperspectral missions where interoperability and uncertainty traceability are paramount.

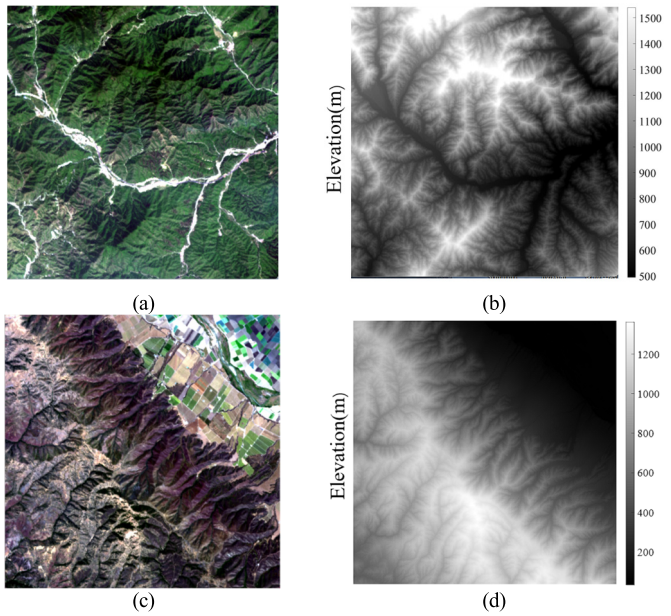



Fig. 3. EnMAP at-sensor radiance composites (641.3, 550.7, 459.0 nm) and corresponding DSMs for (a) and (b) Beijing mountainous region and (c) and (d) California Coast Ranges.

## III. Data and Results

### A. Study Site

Two study areas are selected to evaluate the framework under different terrain and atmospheric conditions. The first site is located in the mountainous region of western Beijing, characterized by terrain relief and dense vegetation cover. The landscape consists primarily of deciduous and coniferous forests, interspersed with exposed rocky outcrops, roads, and sparse built-up areas, making it representative of heterogeneous mountainous environments. Elevation ranges from 494 to 1540 m, with an average slope of 23° and a maximum slope reaching 62°. These conditions make the site particularly suitable for evaluating SR uncertainty in rugged terrain.

The EnMAP Level-1C at-sensor radiance data were acquired on May 28, 2024, under moderate water vapor content but relatively low visibility, representative of typical late-spring atmospheric conditions in northern China. The radiometrically calibrated EnMAP data and the corresponding DSM used for terrain characterization are shown in Fig. 3(a) and (b).

To further verify the robustness of the framework under different environmental conditions, a second study area is selected in California Coast Ranges. The region exhibits distinct geomorphological contrasts: the southwestern portion is dominated by rugged volcanic formations, whereas the northeastern portion consists of relatively flat terrain covered by vegetation and agricultural fields. Elevation ranges from 30 to 1362 m, with an average slope of 16° and a maximum slope of 43°. The EnMAP Level-1C data were acquired on July 17, 2025, under clear coastal conditions with high visibility and humid marine air, representative of typical summer atmospheric conditions in coastal California. The corresponding at-sensor radiance data and DSM are shown in Fig. 3(c) and (d).

EnMAP data were chosen for both sites due to their stable radiometric performance, uncertainty budget, and well-documented calibration framework, which make them particularly suitable for uncertainty quantification and cross-sensor consistency analysis. Nevertheless, the framework is sensor-independent and can be extended to other hyperspectral imaging missions with appropriate radiometric and geometric characterization.

### B. Evaluation of SR Performance

In this section, the accuracy of SR retrieval is evaluated using three approaches: ISOFIT without topographic correction, which is denoted as NC, the Python-based ATCOR (PACO) processor [48], and ISOFIT combined with MM-based topographic correction (denoted as AC + TC). PACO, which serves as the processor for EnMAP Level-2A SR products, shares the same theoretical foundation as ATCOR Software for Airborne and Satellite Imaging [32].

As shown in the zoomed-in subset in Fig. 4, an initial visual inspection indicates that both PACO and AC + TC effectively suppress terrain-induced variations, yielding reflectance maps that are more uniform and topographically neutral compared with the NC results in two study areas.

To quantitatively assess whether terrain effects are adequately removed across the full image, the Pearson correlation coefficient [38] is calculated between the reflectance images and the cosine of the solar incidence angle. This metric is particularly sensitive to changes in direct solar irradiance caused by slope. The results, as shown in Fig. 5(a) for the Beijing mountainous region, demonstrate that both PACO and AC + TC effectively reduce the correlation across most wavelengths, except in the 700–1400 nm range, where an increase in correlation is observed. This anomaly is likely attributable to the strong near-infrared reflectance of vegetation, which remains high even in shaded areas and is, therefore, less

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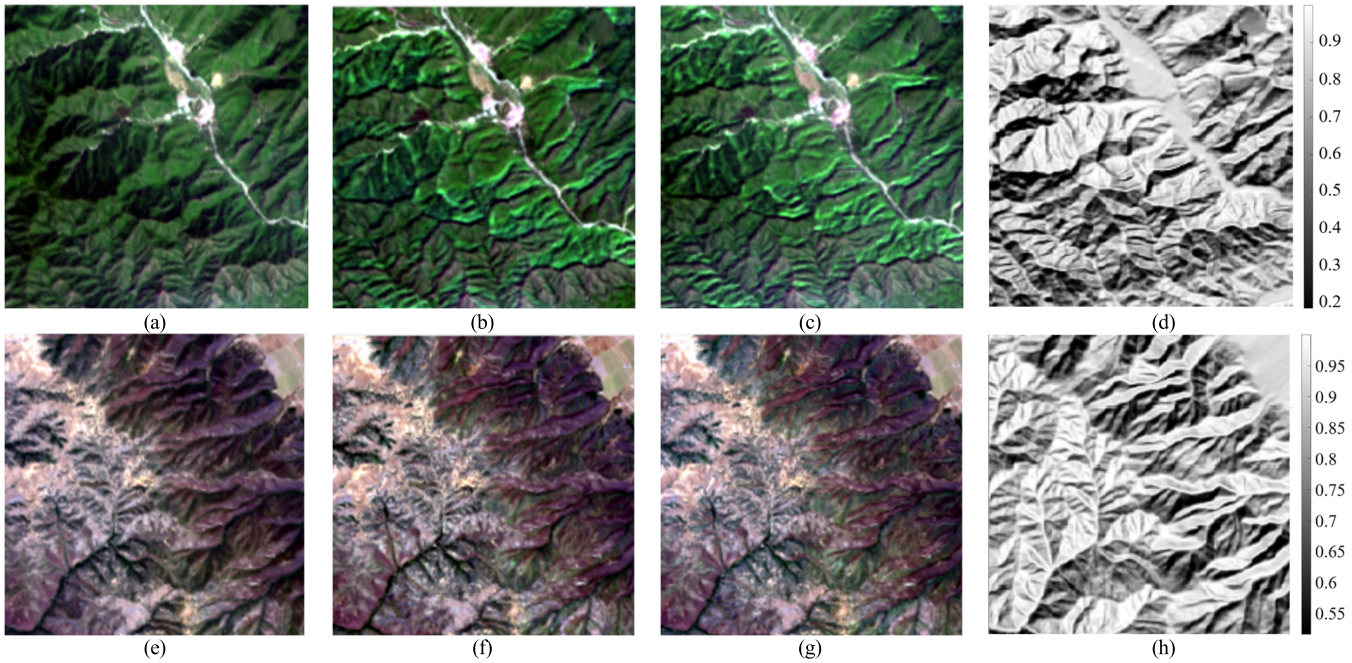



Fig. 4. RGB composite reflectance of a subset of the EnMAP scene (R: 641.3 nm, G: 550.7 nm, B: 459.0 nm) of three methods. (a)–(c) NC, PACO, and AC + TC for the Beijing mountainous area, with (d) showing the corresponding illumination condition ($\cos\beta$), (e)–(g) NC, PACO, and AC + TC results for California Coast Ranges scene, and (h) its corresponding illumination condition ($\cos\beta$).

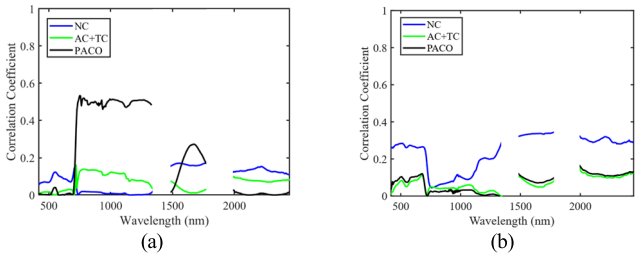



Fig. 5. Correlation coefficients between the whole EnMAP reflectance image and the cosine of the solar incidence angle as a function of wavelength (a) western Beijing mountainous region and (b) California Coast Ranges scene.

affected by illumination geometry. As a result, corrections in this spectral region may lead to suboptimal adjustments. For California Coast Ranges scene [Fig. 5(b)], both AC + TC and PACO consistently suppress terrain-related correlations across the entire 400–2500 nm range, achieving similar performance levels and confirming the robustness of the correction scheme under different topographic and atmospheric conditions.

Overall, the AC + TC and PACO achieve comparable effectiveness in reducing terrain-related correlations.

Although the correction of SR is not the primary focus of this study, establishing its reliability is a necessary prerequisite for subsequent uncertainty analysis. The visual inspection and correlation-based evaluation conducted here provide evidence that the combined atmospheric and topographic correction achieves sufficient radiometric consistency, thereby ensuring that the derived uncertainty estimates are physically meaningful and interpretable. Consequently, this section serves as a validation step to guarantee the robustness of the uncertainty quantification framework.

### C. Uncertainty Quantification

Following the SR uncertainty quantification methodology described in Section II-D, the sources of uncertainty are first quantified, as summarized in Table I. Based on EnMAP radiometric calibration literature, the instrument's radiometric calibration accuracy is adopted as the measurement uncertainty, assumed to follow a Gaussian distribution with a magnitude of 5% ($k = 1$) [49].

For the AW3D30 DSM, both vertical and horizontal accuracy are within 5 m [40], with variation depending on terrain complexity. According to the GUM framework, assuming a Gaussian distribution with a coverage factor of $k = 1.96$ (95% confidence level). Therefore, the corresponding vertical and horizontal uncertainties ($k = 1$) are estimated as $u(p) = 5/1.96$ m and $u(q) = 5/1.96$ m, respectively. The uncertainties of WV, AOT, and SR follow the configuration described in [12]. All quantified uncertainty sources summarized in Table I are expressed as standard uncertainties ($k = 1$) and represent Type B uncertainties, serving as input to the subsequent GUM-based uncertainty propagation analysis.

Based on these uncertainty estimates, the posterior covariance matrix obtained from OE provides the pixel-wise and band-wise uncertainties of the initial SR $u(\rho_t)$. The $u(\rho_t)$ corresponds to the uncertainty of the NC results used in the subsequent comparison. In parallel, the uncertainties of terrain parameters are quantified through the LPU, which propagates DSM-induced uncertainties in slope, aspect, and the solar incidence angle. Finally, by integrating both components into (22), the MM-corrected reflectance and its associated pixel-wise and band-wise uncertainties $u(\rho_{\mathrm{MM}})$ are derived.

The spatial distributions of the SR uncertainties $u(\rho_t)$ and $u(\rho_{\mathrm{MM}})$ for the western Beijing region are shown

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TABLE I
SOURCE AND QUANTIFICATION OF UNCERTAINTY

| Uncertainty Source | Uncertainty Value | Probability Distribution Function |
|---|---|---|
| The instrument measurement noise | 5% | Gaussian |
| DSM vertical uncertainty | 2.55 $m$ | Gaussian |
| DSM horizontal uncertainty | 2.55 $m$ | Gaussian |
| Aerosol optical thickness at 550 nm | 0.1 | Gaussian |
| Water vapor content | 0.5g/cm$^2$ | Gaussian |
| Prior surface reflectance | From clustered and normalized prior spectra in spectral libraries [50]. | Gaussian |

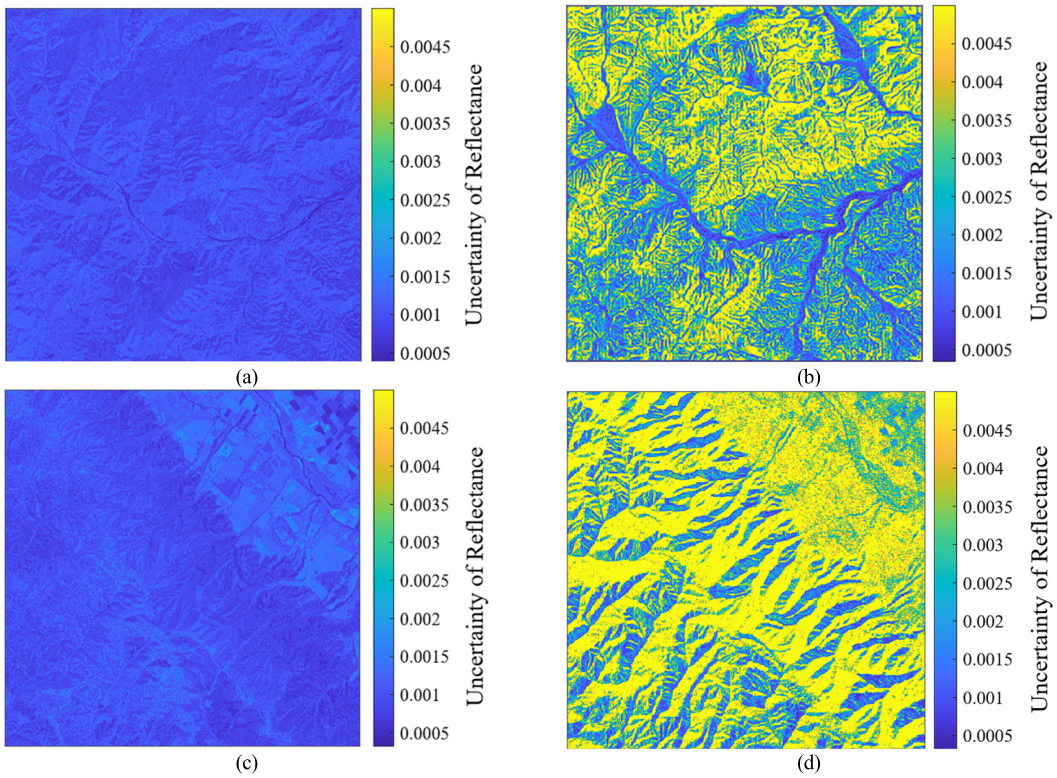



Fig. 6. Spatial distribution of SR uncertainty (the uncertainty of each pixel is calculated as the mean across all valid bands, excluding the absorption bands around 1400 and 1900 nm). (a) NC and (b) AC + TC results for the western Beijing mountainous area. (c) NC and (d) AC + TC results for California Coast Ranges.

in Fig. 6(a) and (b). For visualization purposes, the uncertainty of each pixel is represented as the mean value across all spectral bands, excluding the absorption bands around 1400 and 1900 nm. As illustrated in Fig. 6(b), the overall magnitude of SR uncertainty increases after incorporating terrain effects. Moreover, regions with pronounced terrain variability exhibit noticeably higher uncertainty values, while areas with relatively gentle topography—such as the southeastern portion of the scene—show uncertainty levels comparable to those of the NC results.

For the California Coast Ranges scene [Fig. 6(c) and (d)], a distinct spatial pattern is observed. In Fig. 6(d), sunlit slopes exhibit lower reflectance uncertainties compared to adjacent flat areas. This behavior can be explained by the

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TABLE II
DESCRIPTION OF VALIDATION SITES

| Site ID | Longitude / Latitude (°) | Slope (°) | Aspect (°) | Terrain Condition | Land Cover |
|---|---|---|---|---|---|
| 1 | 115.4846 E / 39.9755 N | 32.35 | 239.34 | Rugged, shaded | Vegetation |
| 2 | 115.4776 E / 39.9749 N | 27.79 | 108.43 | Rugged, sunlit | Vegetation |
| 3 | 115.4804 E / 39.9546 N | 1.62 | 168.69 | Flat | Vegetation |
| 4 | 115.5623 E/ 39.9082 N | 3.18 | 306.86 | Flat | Road |
| 5 | 121.4775 W/ 36.4293 N | 35.52 | 313.45 | Rugged, shaded | Rock |
| 6 | 121.4815 W/ 36.4313 N | 30.55 | 11.41 | Rugged, sunlit | Rock |
| 7 | 121.4624 W/ 36.4498 N | 2.93 | 49.30 | Flat | Vegetation |

local illumination geometry: the slope aspects of these sunlit regions are closely aligned with the solar azimuth direction, resulting in larger cosine values of the solar incidence angle and consequently stronger direct solar irradiance. Under such conditions, the propagated uncertainty derived from the DSM becomes less sensitive to terrain-induced variations, leading to reduced overall reflectance uncertainty. In contrast, the highest reflectance uncertainties are observed on shaded slopes, where lower illumination and weaker signal strength amplify the relative impact of terrain-induced variability.

These spatial patterns highlight that neglecting terrain effects can lead to a systematic underestimation of SR uncertainty. Incorporating terrain-related parameters into the uncertainty propagation process is therefore essential for achieving physically realistic uncertainty estimates and ensuring the interoperability and comparability of SR products across diverse observation geometries.

### D. Validation and Cross-Sensor Consistency Analysis

To validate the derived SR uncertainties from EnMAP, Landsat 8 SR products [51] are selected as a reference due to their widely recognized SI traceability and established uncertainty budget of $0.05\rho + 0.005$. The EnMAP image used for the western Beijing mountainous region was acquired on May 28, 2024, at 11:52:26 local time, while the closest available Landsat scene is captured on May 17, 2024, at 10:58:49 local time. Although there is an 11-day acquisition gap, the spectral differences introduced are considered negligible. This assumption is supported by the spectral stability of the selected land-cover types.

1) *Vegetation:* During late spring in northern China, vegetation phenology is relatively stable, with no rapid green-up or senescence occurring within this short period.
2) *Road:* Impervious surfaces such as asphalt and concrete are spectrally invariant unless affected by transient disturbances (e.g., rain, construction, shadows), none of which were present during the study period.

For the California Coast Ranges scene, the EnMAP image was acquired on July 17, 2025, at 11:30:29 local time, and the corresponding Landsat 8 scene on July 15, 2025, at 10:40:07 local time. Given the two-day acquisition gap, the temporal effects on SR are negligible. The scene features contrasting surface types, including volcanic rock formations, vegetated plains, and agricultural fields, which exhibit high short-term spectral stability under clear-sky summer conditions.

Since the Landsat SR product does not include topographic correction, the same MM method is applied to compensate for terrain-induced illumination effects, and the associated uncertainty is quantified accordingly. For the Landsat SR NC results, the uncertainty is directly derived from the official uncertainty budget ($0.05\rho+0.005$). The subsequent uncertainty evaluation and validation procedures follow the methodology described in Section II-E.

Seven validation sites are selected to ensure representative coverage of different terrain and land cover conditions (Table II). Specifically, Sites 1–4 are located in the mountainous region of western Beijing, while Sites 5–7 are situated in the California Coast Ranges. The sites span both rugged and flat terrain, include slopes under sunlit and shaded illumination, and cover vegetation as well as nonvegetated surfaces such as roads and exposed rock. This strategy provides a balanced sampling of diverse topographic and land cover conditions, thereby enabling a rigorous validation of reflectance and uncertainty consistency between EnMAP and Landsat. To maximize the reduction of uncertainty arising from spatial mismatch, each site is carefully aligned between the two sensors, and a $3 \times 3$ pixel average is used to suppress residual

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TABLE III
QUANTITATIVE COMPARISON OF SR BETWEEN NC AND AC + TC METHODS

| Site ID | *Bias* (NC) | *Bias* (AC+TC) | *RMSE* (NC) | *RMSE* (AC+TC) | $R^2$ (NC) | $R^2$ (AC+TC) |
|---|---|---|---|---|---|---|
| 1 | 0.0033 | 0.0054 | 0.0037 | 0.0089 | 0.9996 | 0.9997 |
| 2 | 0.0102 | 0.0046 | 0.0169 | 0.0070 | 0.9996 | 0.9997 |
| 3 | 0.0033 | 0.0023 | 0.0041 | 0.0032 | 0.9994 | 0.9996 |
| 4 | 0.0143 | 0.0086 | 0.0152 | 0.0090 | 0.9861 | 0.9916 |
| 5 | 0.0201 | 0.0058 | 0.0218 | 0.0071 | 0.9980 | 0.9980 |
| 6 | 0.0195 | 0.0061 | 0.0217 | 0.0068 | 0.9988 | 0.9950 |
| 7 | 0.0250 | 0.0054 | 0.0282 | 0.0063 | 0.9985 | 0.9979 |

sub-pixel geometric offsets. This strategy provides a balanced sampling of diverse topographic and land-cover conditions, thereby enabling a rigorous validation of reflectance and uncertainty consistency between EnMAP and Landsat.

Subsequently, a quantitative comparison between EnMAP and Landsat SR is conducted (Table III). Overall, the AC + TC correction notably improves the agreement between EnMAP and Landsat reflectance, as evidenced by increased $R^2$ values—exceeding 0.99 across all seven validation sites. In terms of Bias and RMSE, the AC + TC results substantially reduce discrepancies between the two sensors at most sites, except for Site 1, where no significant improvement is observed. These quantitative metrics establish a solid foundation for the subsequent analysis of reflectance uncertainty consistency.

The reflectance comparison between EnMAP and Landsat, together with the associated uncertainties, is shown in Fig. 7.

Across all seven sites, applying AC + TC and subsequent uncertainty propagation not only reduces reflectance discrepancies but also improves the overall agreement between the uncertainty-aware reflectance estimates from EnMAP and Landsat. This improvement demonstrates enhanced spectral consistency in both reflectance magnitude and associated uncertainty, particularly in rugged terrain. Notably, at validation sites 2, 5 and 6, all located on sloped areas, substantial divergence between EnMAP and Landsat reflectance is observed in the NC results [Fig. 7(d), (m), and (p)]. After applying AC + TC, the two datasets show markedly improved agreement [Fig. 7(e), (n), and (q)]., confirming that accounting for terrain-induced illumination effects reduces systematic radiometric discrepancies caused by slope and aspect differences.

For sites located in areas with pronounced terrain variation—regardless of whether on shaded or sunlit slopes—the impact of terrain on SR uncertainty is substantial. Neglecting terrain effects leads to underestimation of uncertainty, since slope-induced variations in irradiance generate spatially heterogeneous reflectance responses. Incorporating terrain uncertainty propagation captures this variability, providing a more physically realistic quality assessment.

In relatively flat areas, such as validation site 3 (vegetation), site 4 (road surface), and site 7 (vegetation), the AC + TC also improves the alignment between EnMAP and Landsat reflectance, both in magnitude and spectral shape [Fig. 7(g)–(l) and (s)–(u)].

Furthermore, from the perspective of the $E_n$-ratio, which evaluates consistency between EnMAP and Landsat reflectance, the method consistently reduces $E_n$-ratio values across all seven validation sites. The significant improvements are observed for vegetation and road surfaces, demonstrating enhanced radiometric agreement and uncertainty consistency between the two sensors. These results confirm that the integration of OE- and GUM-based uncertainty quantification with MM topographic correction provides a physically robust framework for assessing SR quality in mountainous terrain.

Nevertheless, as shown in Fig. 7(l), (o), (r), and (u), a residual mismatch remains below 500 nm. This behavior is mainly attributed to the pronounced sensitivity of blue wavelengths to aerosol-retrieval uncertainty [52], [53], as well as a likely underestimation of calibration uncertainty of EnMAP in this spectral region [54]. Because the EnMAP viewing geometry has already been explicitly incorporated into the ATCOR, these deviations are unlikely to be dominated by angular mismatch, but instead reflect the combined influence of aerosol-related and short-wave calibration uncertainties. Future improvements directed at these factors are further discussed in Section IV-C.

Although the uncertainty validation and cross-sensor consistency analysis in this study are demonstrated on two EnMAP

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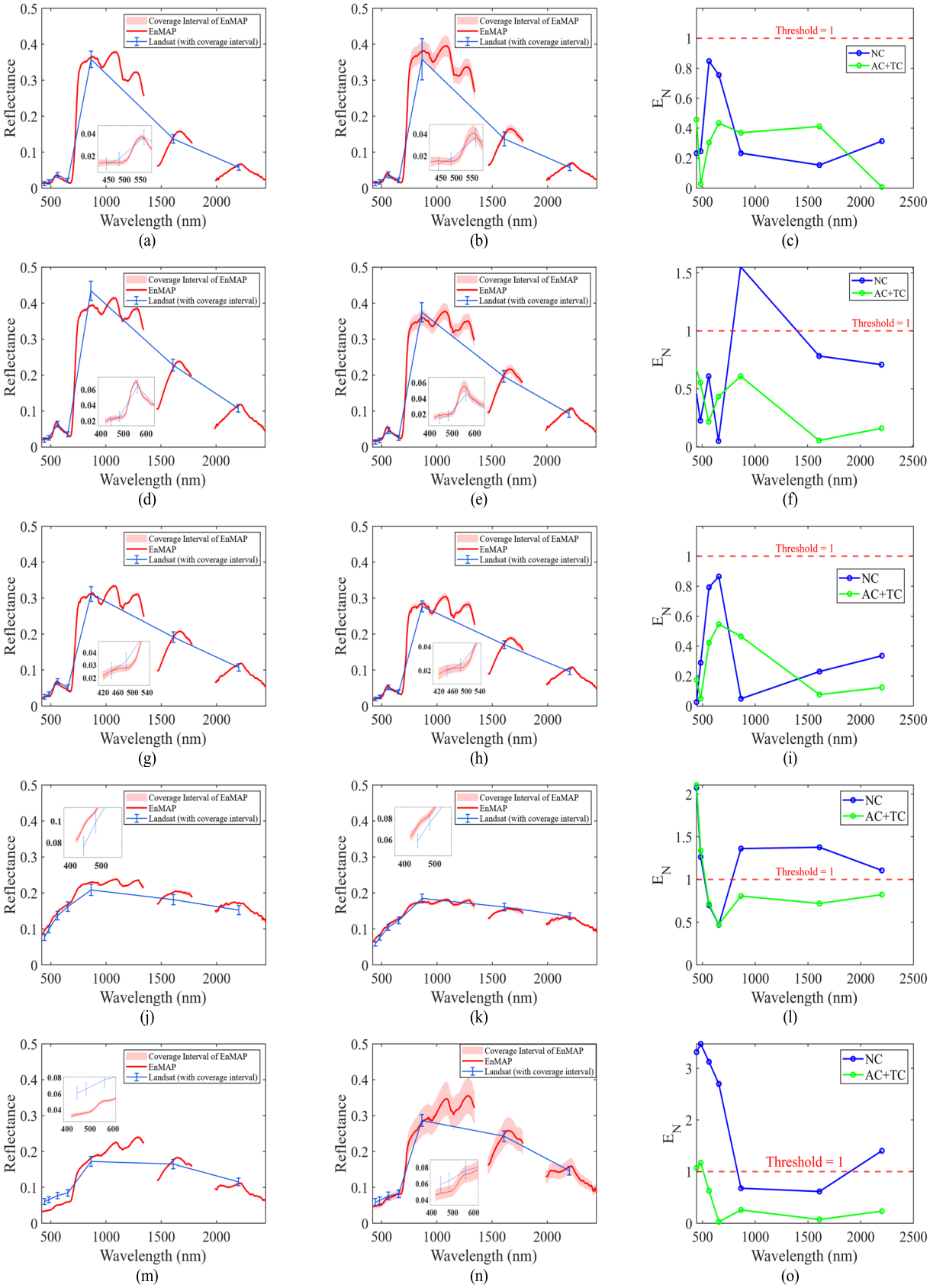



Fig. 7. Validation results for seven selected sites. Panels (a)–(u) correspond to seven validation sites (one per row). (Left) SR and associated coverage intervals from both EnMAP and Landsat under the NC method. (Middle) Same as left, but using AC + TC. (Right) $E_n$-ratio comparison between NC and AC + TC methods.

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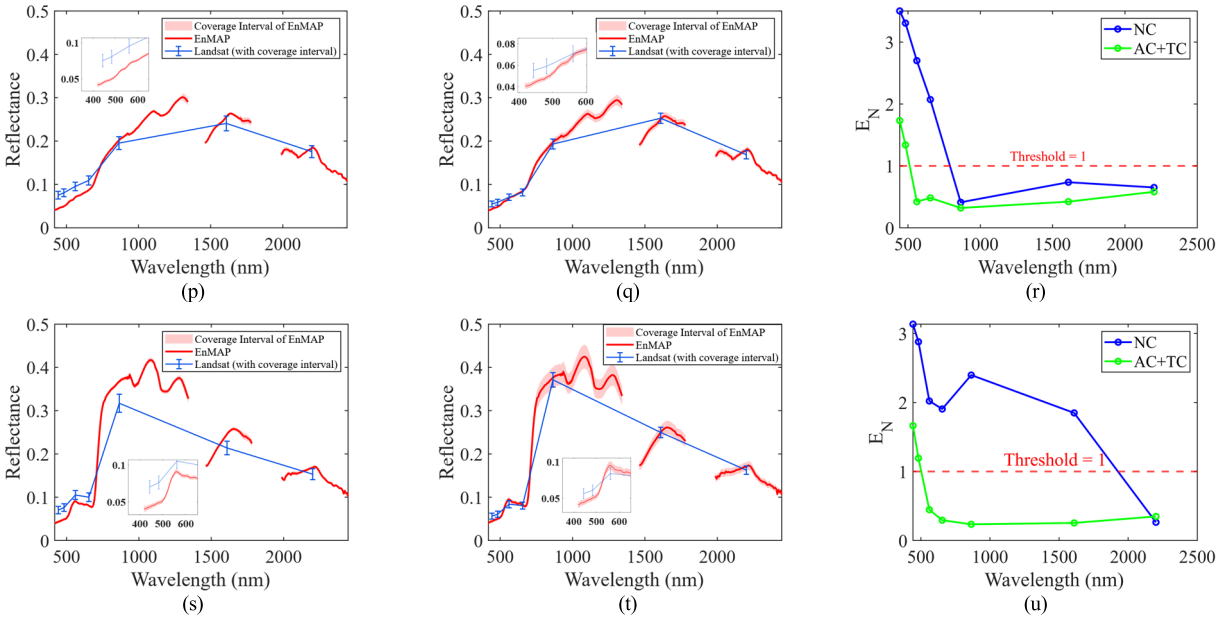



Fig. 7. *(Continued.)*Validation results for seven selected sites. Panels (a)–(u) correspond to seven validation sites (one per row). (Left) SR and associated coverage intervals from both EnMAP and Landsat under the NC method. (Middle) Same as left, but using AC + TC. (Right) $E_n$-ratio comparison between NC and AC + TC methods.

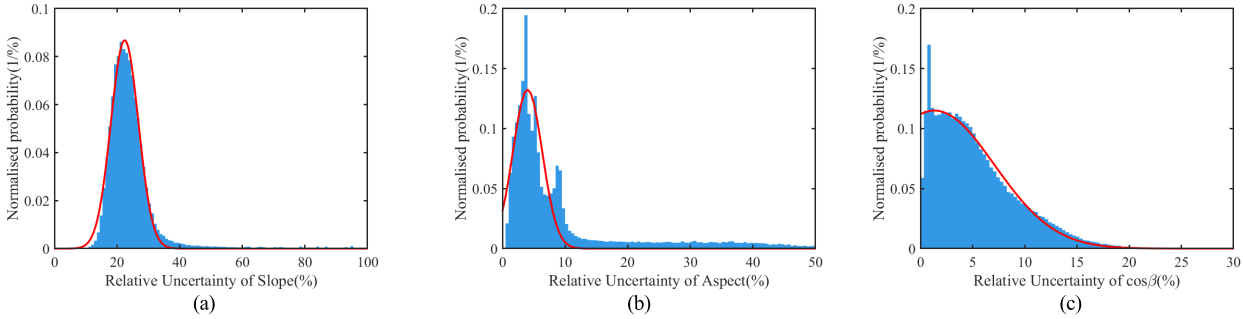



Fig. 8. Histograms of relative uncertainties in (a) slope, (b) aspect, and (c) $\cos\beta$.

scenes, the underlying rationale is general. Uncertainties in terrain parameters, such as slope and aspect, are ubiquitous in hyperspectral imaging over heterogeneous landscapes, and these uncertainties propagate through topographic correction, thereby perturbing the retrieved SR. Since the framework explicitly incorporates uncertainties arising from the sensor noise as well as from atmospheric and topographic correction processes, it can be extended to other scenes and sensors with topographic variations, thereby providing a generalizable approach for uncertainty-based interoperability of hyperspectral products.

## IV. Discussions

### A. Distribution Analysis of Terrain and Reflectance Uncertainty

To investigate how terrain parameter uncertainty influences SR uncertainty, the distribution of slope, aspect, and the solar incidence angle uncertainties are analyzed using the Beijing mountainous region as a representative case, and compared with the distribution of reflectance uncertainty across six typical spectral bands, thereby characterizing the statistical distribution of SR uncertainty.

Histograms of the relative uncertainties in slope, aspect, and cosine of the solar incidence angle are generated for all pixels in the full EnMAP scene as shown in Fig. 8. Relative uncertainty is defined as the ratio of the standard uncertainty to the parameter value (i.e., $u(a)/a \times 100\%$), allowing for a normalized comparison across parameters with different magnitudes. Fig. 8 reveals several key observations.

1) *Relative Uncertainty of Slope:* The relative uncertainty in slope for most pixels falls within the range of 10%–40%.
   The distribution closely follows a Gaussian shape, exhibiting a clear unimodal normal distribution with good fitting performance.
2) *Relative Uncertainty of Aspect:* The distribution of aspect uncertainty is strongly skewed, with a large number of pixels concentrated below 10%, but with a long tail extending beyond 40%. The Gaussian fitting

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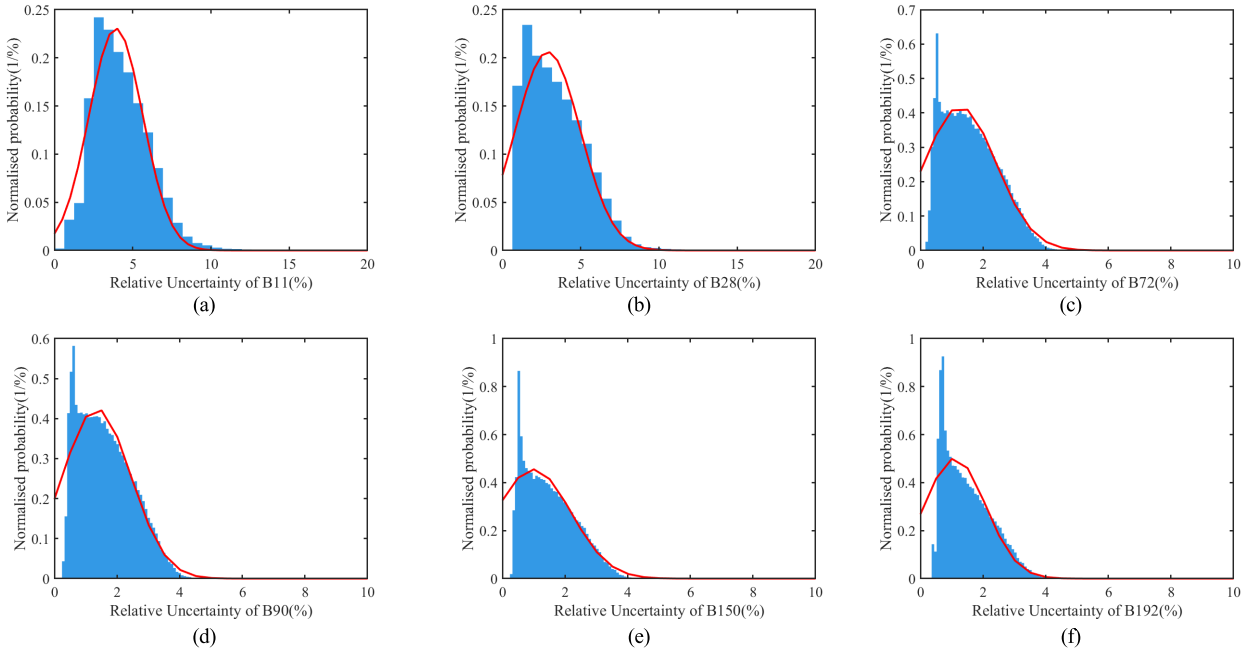



Fig. 9. Histograms of relative uncertainties in different bands of reflectance (a) B11, (b) B28, (c) B72, (d) B90, (e) B150, and (f) B192.

performs poorly, suggesting that the uncertainty in the aspect may not follow a standard normal distribution.

3) *Relative Uncertainty of* $\cos\beta$*:* This parameter integrates the effects of slope, aspect, and solar elevation, directly impacting irradiance estimation in topographic correction. Its distribution is also skewed but primarily concentrated below 20%, and the Gaussian fit performs reasonably well.

To investigate the distributional characteristics of SR uncertainty, six spectral bands spanning the EnMAP range are analyzed: B11 (468.41 nm), B28 (550.69 nm), B72 (839.98 nm), B90 (944.47 nm), B150 (1609.02 nm), and B192 (2191.01 nm). Their reflectance uncertainty distributions are shown in Fig. 9. The results indicate that the magnitude of reflectance uncertainty varies across spectral regions. Nevertheless, despite these differences, the distributions remain compact and are well approximated by Gaussian functions.

Nevertheless, certain uncertainty sources—such as long-tailed retrieval errors in aerosol and water vapor estimation—may deviate from Gaussian assumptions and exhibit heteroscedastic behavior. Such deviations can lead to heavy-tailed or asymmetric uncertainty propagation. If non-Gaussian uncertainty distributions occur, the assumption of a fixed coverage factor (e.g., $k = 1.96$ for 95% confidence) in (26) would no longer hold. In such cases, the $E_n$-ratio can be generalized as follows:

$$E'_N = \frac{|\rho_1 - \rho_2|}{k_p \cdot \sqrt{u_1^2 + u_2^2}} \tag{27}$$

where $k_p$ is an empirical coverage factor derived from the observed uncertainty distribution using nonparametric approaches such as bootstrap resampling or quantile-based estimation. This adaptation allows the framework to remain valid under non-Gaussian or heteroscedastic conditions, ensuring more robust uncertainty-based consistency assessment.

Overall, across all six bands, the relative uncertainty distributions of SR are well approximated by Gaussian functions in this study, which justifies the use of the $E_n$-ratio in (26) under the Gaussian assumption.

### B. Terrain–Reflectance Uncertainty Relationship

Building upon the distribution analysis of terrain parameter uncertainties, this section investigates the intricate relationship between the absolute uncertainties of slope and aspect and those of SR using the Beijing scene as a representative example. Heatmap scatter plots for six representative spectral bands (Fig. 10), consistent with earlier analyses, illustrate the statistical relationship between terrain and reflectance uncertainties.

A clear and consistent trend emerges: as the absolute uncertainty in slope increases, the uncertainty in SR correspondingly rises. This is visually reflected by the color gradient in the heatmaps, which transitions smoothly from deep red (indicating lower reflectance uncertainty) to yellow (indicating higher uncertainty). Such a pattern strongly suggests that areas with more uncertain slope measurements tend to exhibit larger variability in reflectance retrievals, likely because slope affects the angle of solar incidence and the local illumination geometry, which are critical factors in topographic correction algorithms. In contrast, the impact of absolute aspect uncertainty on SR uncertainty appears considerably less significant. The heatmaps show that changes in aspect uncertainty do not correspond to pronounced variations in reflectance uncertainty, as indicated by relatively uniform color distributions across different aspect uncertainty levels. This finding implies that while aspect—representing the compass

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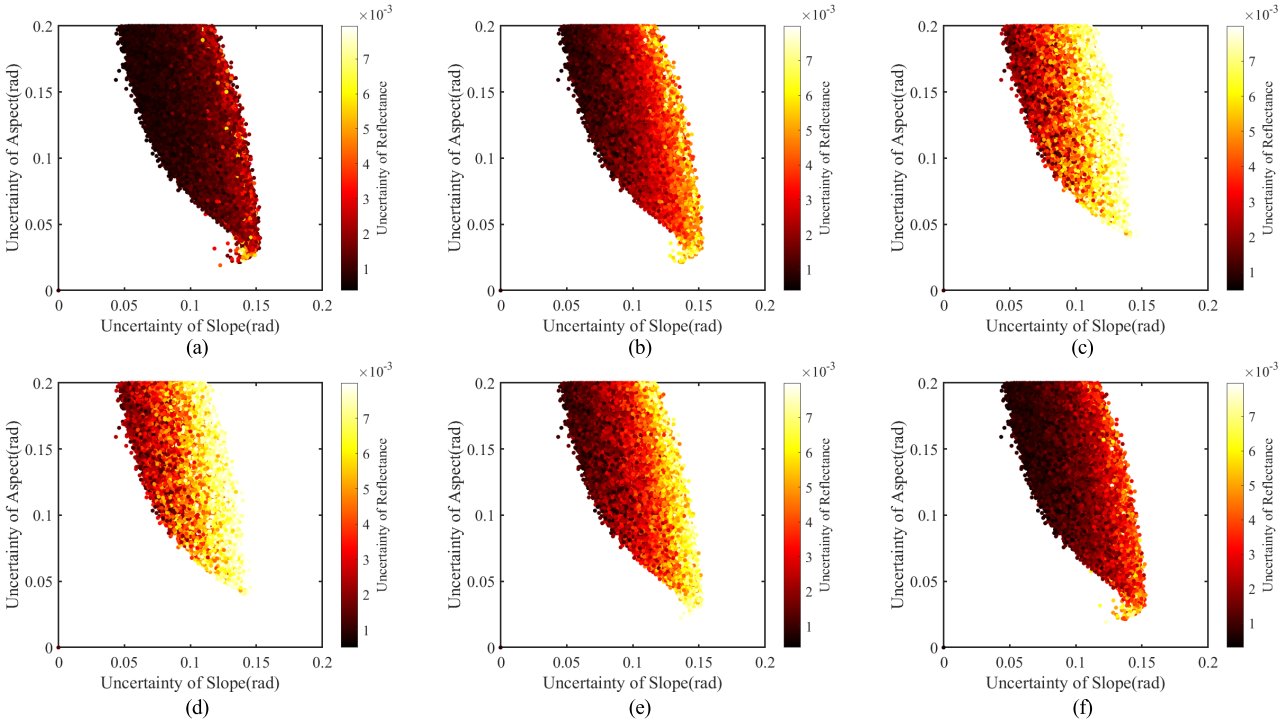



Fig. 10. Heatmap scatter plots showing the relationship between absolute uncertainties in slope/aspect and SR (a) B11, (b) B28, (c) B72, (d) B90, (e) B150, and (f) B192.

direction of the slope—does influence illumination conditions, its uncertainty exerts a weaker effect on the overall uncertainty of reflectance inversion within the EnMAP topographic correction framework. Consequently, this analysis highlights slope uncertainty as the dominant terrain-related contributor to reflectance uncertainty, emphasizing the need for more accurate slope measurements or modeling in topographic correction processes. Meanwhile, aspect uncertainty, though not negligible, plays a comparatively minor role and may be considered a secondary factor in uncertainty propagation for reflectance retrievals. This weaker influence can be explained by the relatively small sensitivity of $\cos\beta$ to the aspect, particularly under near-noon illumination conditions.

### C. Future Work

Although this study considers the impact of terrain parameters on hyperspectral SR uncertainty and validates the uncertainty estimates through cross-comparison with Landsat 8 SR products, it should be emphasized that the MM method employed here remains a post hoc topographic correction approach. The MM method primarily accounts for the influence of the cosine of the solar incidence angle, while neglecting reflectance contributions from adjacent slope and neighboring objects (i.e., adjacency effects and topographic mutual illumination) [55]. In addition, the MM method empirically accounts for surface anisotropy but lacks explicit BRDF modeling. Its computational simplicity makes it well-suited for operational use; however, future research should move toward a physically integrated correction framework that simultaneously models atmospheric, topographic, and BRDF effects [56] along with uncertainty quantification.

Moreover, the analysis in Section III-D indicates that short-wavelength residuals—particularly the cases where the $E_n$-ratio exceeds 1 below 500 nm—stem from the pronounced sensitivity of these bands to aerosol-retrieval uncertainty as well as from a likely underestimation of calibration uncertainty of EnMAP. These findings highlight the need to refine the specification of prior aerosol uncertainty [57] and to incorporate wavelength-dependent radiometric calibration uncertainty more rigorously. While viewing-geometry mismatch is found to be a minor factor in this study, future work should also evaluate its impact under more complex angular configurations or when cross-sensor temporal separation is larger. Addressing these sources of uncertainty jointly within a unified radiative-transfer and uncertainty-propagation framework will further enhance the robustness of cross-sensor consistency analysis.

In addition, extending the uncertainty-aware framework to multitemporal and multisensor analyses represents another important direction. This includes investigating how temporal changes in vegetation phenology or surface conditions interact with terrain-induced effects, as well as testing interoperability with forthcoming hyperspectral missions such as the Copernicus Hyperspectral Imaging Mission for the environment (CHIME) [58] and hyperspectral infrared imager (HyspIRI) [59]. Such developments will advance toward a fully uncertainty-aware characterization of SR that is not only operationally practical but also traceable and comparable across different sensors, times, and regions.

## V. Conclusion

This study presents a novel and computationally efficient framework for SR uncertainty quantification in hyperspectral

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remote sensing, achieved by combining ISOFIT-based ATCOR with an MM approach and extending it through OE–GUM uncertainty propagation. Unlike conventional methods that focus solely on correction accuracy, the framework emphasizes uncertainty as a quantitative indicator of data quality, explicitly accounting for terrain-induced effects in mountainous regions.

Validation through cross-comparison with Landsat 8 SR products across two study areas—representing distinct terrain and atmospheric conditions—demonstrates that the derived uncertainties provide a robust measure of intersensor radiometric consistency, offering new insights into the comparability of hyperspectral and multispectral data sources. Statistical analyses further reveal that reflectance uncertainty generally follows a Gaussian distribution and that slope uncertainty exerts the strongest influence on reflectance variability.

Overall, the method shifts the focus from correction performance to uncertainty-aware evaluation, establishing a practical pathway for harmonizing hyperspectral products across sensors and contributing to the uncertainty characterization requirements of upcoming global missions such as CHIME and HysPIRI.

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